

Concepts and Categorization

Theoretical Descriptions of the Nature of Concepts

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Forming New Concepts and Classifying New Instances

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C H A P T E R

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If your college or university is like the one where I teach, you probably have to fulfill certain graduation requirements—among them, distribution requirements, which mandate your taking a certain number of courses in each of several groups. For example, Carleton College has four distribution groups: arts and literature (including most courses in studio art, art history, English, literature in translation, literature in foreign languages, and music), social sciences (including educational studies, economics, political science, psychology, and sociology/anthropology), natural science and mathematics (including astronomy, biology, computer science, chemistry, geology, mathematics, and physics), and humanities (including history, philosophy, and religion). The groupings of subject matter into larger distribution groups illustrates my college's categorization, or assignment of courses to groupings.

Of course, not all colleges have the same groups or the same assignment of courses to groups. For example, at other schools my home discipline, psychology, is

often assigned to the natural sciences division. At many schools, arts and humanities are grouped together. I'm not entirely sure how the Carleton grouping came to be, but I am sure the dean or committee that created it had a mental representation of this category, something that a cognitive psychologist would call a *concept*.

We have encountered the idea of mental representations several times before. We have seen that many cognitive psychologists (particularly those within the information-processing paradigm) believe that such representations guide cognitive processing and behavior. How you regard something may often be influenced by what type of thing you believe it to be. For example, you would probably react and behave one way if told that a severe thunderstorm was taking place outside, and another way if you were told it was a hurricane. Your classification of storms into two categories suggests that you see distinctions between them. Presumably, such distinctions cause your reactions to storms to depend on classification.

A related, real-life example comes from medical diagnosis. Suppose you wake up one day feeling achy, lethargic, congested, and feverish. Your symptoms could indicate nothing more serious than flu. Or your symptoms could be the harbinger of a much more serious illness. It is your doctor's job to make the diagnosis, which essentially is to assign your pattern of symptoms to a category corresponding to known diseases or medical problems. The categorization allows the physician to determine appropriate treatment and predict the time course of recovery. To make the diagnosis, your physician must have an idea of the various categories (possible medical problems) to be considered. Indeed, physicians are not the only ones who categorize illnesses, as shown by a recent study of laypeople's categorization of forms of mental illness (Kim & Ahn, 2002).

In this chapter, we'll look at concepts and how they are formed. We'll examine different theoretical descriptions of how concepts are structured and their implications for how we assume our mental representations work. We'll then focus on how concepts are accessed and used in categorizing new objects, patterns, or events. Many ideas discussed in the early part of the chapter will extend and elaborate on proposals presented in Chapter 3, "Perceiving Objects and Recognizing Patterns" (pattern recognition and classification have many similarities, as we shall see), and in Chapter 7, "Memory for General Knowledge." Similarly, our examination of categorization will anticipate some later discussions about language, thinking, reasoning, and decision making (Chapters 10–13). You can probably already see that an understanding of how people form and use concepts is relevant to several other cognitive processes and abilities. Medin (1989) has in fact argued that "concepts and categories serve as building blocks for human thought and behavior" (p. 1469). Lamberts and Shanks (1997) have argued that the issue of how things such as concepts are mentally represented is a central concern of cognitive psychology.

What are concepts and categories, and how do they differ? The distinction turns out to be a little blurry but can still be made. Medin (1989) defined a concept as “an idea that includes all that is characteristically associated with it” (p. 1469). In other words, a **concept** is a mental representation of some object, event, or pattern that has stored in it much of the knowledge typically thought relevant to that object, event, or pattern. Most people’s concept of “dog,” for example, would include information to the effect that a dog is an animal, has four legs and a tail, has a reputation as “man’s best friend,” is a common pet, and so on.

A **category** can be defined as a class of similar things (objects or entities) that share one of two things: either an essential core (example: why all science courses are considered “science”) or some similarity in perceptual, biological, or functional properties (Lin & Murphy, 2001). When a psychologist thinks about categories, she usually thinks about several different ones, into which various things get sorted. In the game 20 Questions, a common opener is, “Is it an animal, vegetable, or mineral?” This question seeks to categorize, or sort, the to-be-guessed item into one of three things. Sometimes categories are described as existing objectively in the world, and concepts are described as mental representations of categories (Medin, 1989).

Concepts help us establish order in our knowledge base (Medin & Smith, 1984). Concepts also allow us to categorize, giving us mental “buckets” in which to sort the things we encounter, letting us treat new, never-before-encountered things in the same way we treat familiar things that we perceive to be in the same set (Neisser, 1987). Categorization also allows us to make predictions and act accordingly. If I see a four-legged creature with a tail coming toward me, my classification of it as either a dog or a wolf has implications for whether I’ll want to call to it, run away, pet it, or call for help. Smith and Medin (1981) elaborated on the important role concepts play in our mental life:

Without concepts, mental life would be chaotic. If we perceived each entity as unique, we would be overwhelmed by the sheer diversity of what we experience and unable to remember more than a minute fraction of what we encounter. And if each individual entity needed a distinct name, our language would be staggeringly complex and communication virtually impossible. Fortunately, though, we do not perceive, remember, and talk about each object and event as unique, but rather as an instance of a class or concept that we already know something about. (p. 1)

We will first examine different theoretical accounts about the nature and structure of concepts. Next, we’ll look at how concepts are formed or acquired. Finally, we’ll examine how people actually use concepts through the process of categorization. Throughout this chapter, we’ll be focusing on concepts of objects and nouns, because they are the most commonly studied at present in

cognitive psychology. We will see, however, that the kind of concept studied may affect the concept theories that are subsequently created. So it will be useful to keep in mind that psychologists have yet to explore fully the entire range of people's concepts.

■ THEORETICAL DESCRIPTIONS OF THE NATURE OF CONCEPTS

In Chapter 7, we reviewed proposals for how our knowledge bases are represented and organized. Models of semantic memory describe the ways in which representations of different concepts are interrelated. Here we will concentrate on the representation and organization of individual concepts. We will explore five distinct proposals on how concepts are represented and structured. Each one provides a different answer to the question “What information do we have when we have a particular concept?” Each proposal will therefore have different implications for the question of how concepts are formed, acquired, or learned.

The Classical View

The classical view of concepts was the dominant view in psychology up until the 1970s and dates back to Aristotle (Smith & Medin, 1981). This proposal is organized around the belief that all examples or instances of a concept share fundamental characteristics, or **features** (Medin, 1989). In particular, the **classical view of concepts** holds that the features represented are individually necessary and collectively sufficient (Medin, 1989). To say a feature is individually necessary is to say that each example must have the feature if it is to be regarded as a member of the concept. For example, “has three sides” is a necessary feature of the concept *triangle*; things that do not have three sides are automatically disqualified from being triangles. To say that a set of features is collectively sufficient is to say that anything with each feature in the set is automatically an instance of the concept. For example, the set of features “has three sides” and “closed, geometric figure” is sufficient to specify a triangle; anything that has both is a triangle. Table 8-1 presents some other examples of sets of features or of concepts that are individually necessary and collectively sufficient.

The classical view of concepts has several implications. First, it assumes that concepts mentally represent lists of features. That is, concepts are not representations of specific examples but rather abstractions containing information about properties and characteristics that all examples must have. Second, it assumes that membership in a category is clear-cut: Either something

Table 8-1 Examples of Concepts and Their Features

	<i>Bachelor</i>	<i>Triangle</i>	<i>Uncle</i>	<i>Prime Number</i>
Concept	Male	Geometric figure	Male	Integer divisible by two numbers: itself and 1
Features	Adult Unmarried Human	Three-sided Planar	Sibling One or more siblings has a child	

has all the necessary and sufficient features (in which case it is a member of the category), or it lacks one or more of the features (in which case it is not a member). Third, it implies that all members within a category are created equal: There is no such thing as a “better” or “worse” triangle.

Work by Eleanor Rosch and colleagues (Rosch, 1973; Rosch & Mervis, 1975) confronted and severely weakened the attraction of the classical view. Rosch found that people judged different members of a category as varying in “goodness.” For instance, most people in North America consider a robin and a sparrow very good examples of a bird but find other examples, such as chickens, penguins, and ostriches, not as good. Notice the problem this result presents for the classical view of concepts. The classical view holds that membership in a category is all-or-none: Either an instance (such as robin or ostrich) belongs to a category, or it doesn’t. This view has no way to explain people’s intuitions that some birds are “birdier” than others.

People’s judgments of typicality, the “goodness” of the instance in the category, was later shown to predict several aspects of their performance on different tasks. For example, participants in a sentence verification task were faster to respond (true or false) to a sentence such as “A robin is a bird” than to a sentence such as “A chicken is a bird” (McCloskey & Glucksberg, 1978; Rosch, 1973; Smith et al., 1974). Asked to list instances of a concept, people were more likely to list typical than atypical instances (Mervis, Catlin, & Rosch, 1976). In semantic priming studies (see Chapter 7 for a review), highly typical instances often led to better priming (Rosch & Mervis, 1975; Rosch, Simpson, & Miller, 1976).

All these results are not easily explained within a classical framework. In addition, other studies cast doubt on the idea that people typically store and refer to a list of necessary features when judging category membership. McCloskey and Glucksberg (1978) gave participants a list of items and asked them to judge whether the items belonged to certain categories (for example,

“Does ‘chair’ belong to the category ‘furniture?’”). The classical view would predict very strong agreement across people, but McCloskey and Glucksberg’s participants in fact disagreed considerably on atypical instances (for example, “Do ‘bookends’ belong to the category ‘furniture?’”). Participants were often inconsistent in their own responses in different sessions. This result argued especially strongly against the classical assumption that categories have clearly defined boundaries. Finally, even when given specific instructions to do so, most people cannot generate lists of features that are individually necessary and collectively sufficient to specify membership in a category (Ashcraft, 1978; Rosch & Mervis, 1975).

The Prototype View

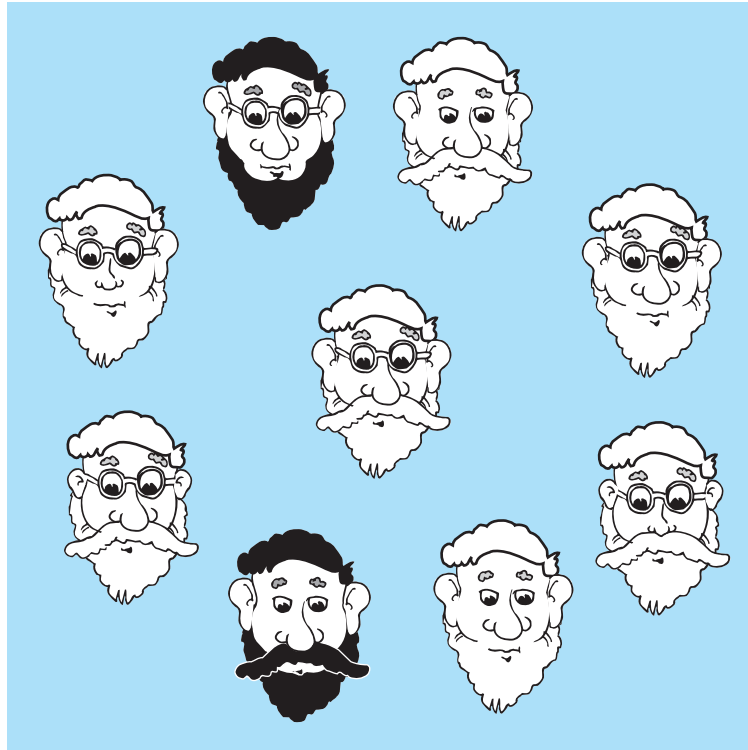
A second theoretical view of the nature of concepts, known as the *prototype view*, was proposed in the 1970s. The **prototype view of concepts** denies the existence of necessary-and-sufficient feature lists (except for a limited number of concepts such as mathematical ones), instead regarding concepts as a different sort of abstraction (Medin & Smith, 1984). Like perceptual researchers (see Chapter 3), conceptual researchers believe in the existence of mental **prototypes**, idealized representations of some class of objects or events. Specifically, researchers studying the prototype view of concepts hold that prototypes of concepts include features or aspects that are *characteristic*—that is, typical—of members of the category rather than necessary and sufficient. No individual feature or aspect (except very trivial ones, such as “is an object”) need be present in the instance for it to count as a member of the category, but the more characteristic features or aspects an instance has, the more likely it is to be regarded as a member of the category.

The prototype view of concepts and categories often refers to the **family resemblance structure of concepts** (Wittgenstein, 1953), a structure in which each member has a number of features, sharing different features with different members. Few, if any, features are shared by every single member of the category; however, the more features a member possesses, the more typical it is. Figure 8-1 provides an example of family resemblance. Note that the Smith brothers (modeled after the men on Smith Bros. Cough Drop boxes) have several shared features: light hair, bushy mustache, large ears, and eyeglasses. Not every Smith brother has every feature, but the brother in the middle, having them all, would likely be judged by Smith friends to be the most typical Smith of the bunch. Note that he shares big ears, eyeglasses, and light hair with the brother in the “ten o’clock” position and a mustache and big ears with the “seven o’clock” brother. Indeed, different pairs of brothers share different features.

The prototype view of concepts explains typicality effects by reference to family resemblance. The idea is that the more characteristic features an instance

FIGURE 8-1 ■ *An example of family resemblance.*

SOURCE: Armstrong et al. (1983, p. 269).



of a concept has, the stronger the family resemblance between that instance and other instances, and therefore the more typical an instance it is. Presumably, then, a robin is thought of as a more typical bird than a penguin because the robin possesses more characteristic bird features, such as “is small,” “flies,” “eats worms,” and “lives in a tree.” Even with well-defined concepts such as *bachelor*, some examples seem more bachelorlike than others.

For example, is my 13-year-old son Tim a good example of a bachelor? He is male and unmarried. And probably, he’s a better example of a bachelor today than he was 10 years ago. What about the pope? The point here is that both people may meet the technical definition of a bachelor (there’s some disagreement over whether the definition includes “adult”), but neither is as good an example as might be someone such as the current male teenage heartthrob.

In one set of studies, Rosch and Mervis (1975) presented their undergraduate participants with terms (such as *chair*, *car*, *orange*, *shirt*, *gun*, *peas*) from six different superordinate categories (such as “furniture,” “vehicle,” “fruit,” “clothing,” “weapon,” “vegetable”) and asked them to list attributes “common to and

characteristic of” those objects. So, for example, for the word *chair* a participant might list “has four legs; used to sit in; sometimes has arms; used in homes and offices.” Then Rosch and Mervis tallied a list of all the attributes any participant listed for all basic-level terms belonging to a superordinate category (for example, all the attributes listed for *chair*, *sofa*, *table*, *dresser*, *desk*, *bed*, *clock*, *closet*, *vase*, *telephone*). Next, they computed, for each item, the number of attributes commonly listed for it. They found that items such as chair and sofa—ones that seem more prototypical of the superordinate category “furniture”—had many more of the “furniture” attributes listed than did items such as clock or telephone, which are both not at all prototypical examples of furniture. However, very few (0 or 1) attributes in any of the six superordinate categories were true of all 20 items for the category (for example, attributes true of all fruits).

A prototype, then, is some sort of abstraction that includes all the characteristic features of a category. The prototype may or may not be an actual instance of the category. Prototypes are often thought of as mental “summaries” or “averages” of all the instances, although there are some problems with this view (Barsalou, 1985). The general idea of the prototype view, then, is that concepts have one or more “core” representations, based on a family resemblance structure, but have no rigid boundaries.

Rosch and her colleagues (Rosch, Mervis, Gray, Johnson, & Boyes-Braem, 1976) made another important discovery about concepts. Although concepts exist at many different levels of a hierarchy (for example, “Bernese mountain dog,” “dog,” “canine,” “mammal,” “animal”), one level of abstraction appears psychologically fundamental. They called this the “basic” level and distinguished it from both higher-level (superordinate) and lower-level (subordinate) concepts.

To understand the distinctions between the **basic level of categorization** and other levels, consider the purpose of categorization. On the one hand, we want to group together similar objects, events, people, ideas, and so on. On the other hand, we want our categorization to distinguish among objects, events, people, and ideas that differ in important ways. There must be some compromise between these two goals. Rosch and colleagues consider the basic level to be the best compromise.

“Piano” and “guitar” are examples of two basic-level categories. Such categories include members that are maximally similar to one another, unlike **superordinate levels of categories** (such as “musical instruments”), which contain members (such as pianos and guitars) that are dissimilar in several respects. At the same time, basic-level categories are most differentiated from one another, especially relative to subordinate categories. “Grand piano” and “upright piano” are two categories at the **subordinate level of categories**; these categories are less distinct than are two basic-level categories, such as “piano” and “guitar.” The list in Table 8-2 presents examples of basic-level categories, along with related superordinate and subordinate categories.

Table 8-2 Basic-Level Categories with Related Superordinate and Subordinate Categories

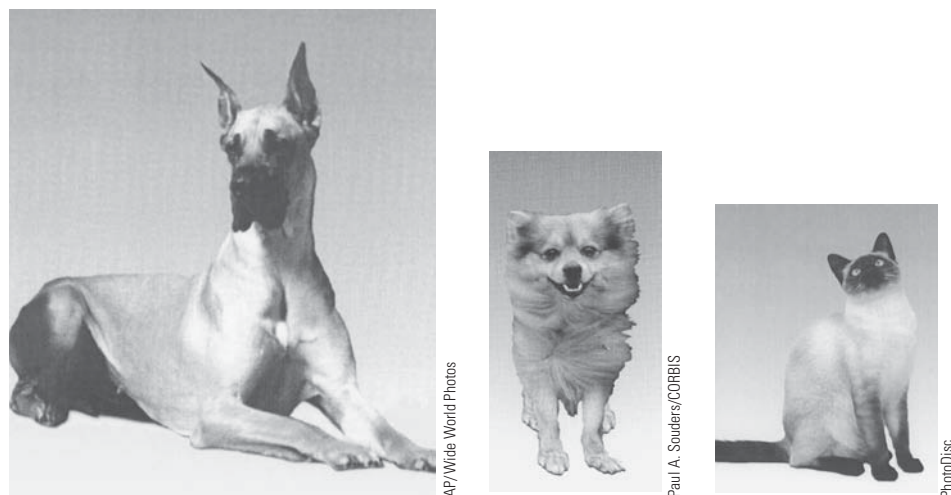
<i>Superordinate</i>	<i>Basic Level</i>	<i>Subordinate</i>
Musical instrument	Guitar	Classical guitar Folk guitar
	Piano	Grand piano Upright piano
	Drum	Bass drum Kettle drum
	Fruit	Delicious apple McIntosh apple
	Peach	Cling peach Freestone peach
Tool	Grapes	Concord grapes Green seedless grapes
	Hammer	Claw hammer Ball-peen hammer
	Saw	Hack handsaw Cross-cutting handsaw
	Screwdriver	Phillips screwdriver Regular screwdriver
	Clothing	Levis Double-knit pants
Furniture	Socks	Knee socks Ankle socks
	Shirt	Dress shirt Knit shirt
	Table	Kitchen table Dining room table
	Lamp	Floor lamp Desk lamp
	Chair	Kitchen chair Living room chair
Vehicle	Car	Sports car Four-door sedan
	Bus	City bus Cross-country bus
	Truck	Pickup truck Tractor-trailer truck

SOURCE: Rosch, Mervis, et al. (1976, p. 388).

The prototype view does a very good job at explaining why certain members of a category are seen as more typical than others. It also explains why people have a hard time providing strict definitions of their concepts: Strict definitions do not exist. Finally, the prototype view can explain why some classifications are especially easy to make and others are unclear. Take tomatoes, which some people classify as a vegetable and others classify as a fruit. Tomatoes are often eaten with other vegetables instead of with other fruits, and they share some similarities with other vegetables. However, to a biologist, tomatoes are a fruit because they develop from the flower of the plant (technically, the pistil). Vegetables, in contrast, are any nonreproductive parts of a plant, such as the stem or root. The prototype view explains the ambiguity of tomatoes: They share features both with vegetables (leading to classification as a vegetable) and with fruits (leading to classification as a fruit).

The prototype view is not wholly free of problems. For one thing, it fails to capture people's knowledge about the limits of conceptual boundaries. To illustrate, even though a Pomeranian seems in many ways more similar to a Siamese cat than to a Great Dane, the Pomeranian and Great Dane are classified together as dogs (Komatsu, 1992). The prototype view has a hard time telling us why. Unlike the classical view, which sets constraints or boundaries around which things can and can't belong to a category, the prototype view does not specify clear constraints.

Rosch and colleagues (Rosch, 1973; Rosch & Mervis, 1975; Rosch, Mervis, et al., 1976) have argued that some constraints around different categories come



A Great Dane, a Pomeranian, and a Siamese cat: Even though the overall similarity may be greater between the latter two, the former two are classified together in the category “dogs.”

from the environment itself. Having wings and being able to fly, for example, tend to co-occur, often in those things we call *birds* (but also in airplanes, butterflies, and insects). Boundaries between categories, then, come not just from us as cognitive processors of information but from the way the world works: Certain patterns of attributes or features occur in the world, and others don't (Komatsu, 1992; Neisser, 1987). People's main job in categorizing, then, is to pick up information about the world's regularities, not to impose arbitrary groupings, as the classical view might imply. (The idea of "picking up information" about the world might remind the alert student of Gibsonian theories of perception, discussed in Chapter 3.)

A second problem for the prototype view has to do with typicality ratings. Barsalou (1985, 1987) and Roth and Shoben (1983) showed that the typicality of an instance depends to some extent on context. So although a robin may be seen as a typical bird in the context of birds you see in the neighborhood, it is atypical of birds you see in a barnyard. These findings contrast with the idea that a member of a category has a certain level of typicality. Instead, typicality apparently varies with the way the concept itself is being thought about.

Studies by Armstrong, Gleitman, and Gleitman (1983) demonstrated additional problems with typicality ratings. In these studies, the investigators asked participants to rate the typicality of instances of both natural concepts (such as "vehicle" or "fruit") previously studied by Rosch and her colleagues and of well-defined concepts (such as "even number," "female," "geometric figure"). Armstrong et al. found that participants happily rated the typicality of members of well-defined categories, generally agreeing that 3 is a more typical odd number than 57, for example. The same participants also agreed, however, that the category "odd number" was well defined and that it makes little sense to talk about degree of membership in the category: Numbers either are or are not odd. The investigators concluded that the typicality ratings task is flawed, at least for discovering the underlying representation of concepts.

The Exemplar View

The previous two views of concepts both hold that concepts are some sort of mental abstraction or summary. In other words, individual instances are not specifically stored or mentally represented but instead are averaged into some sort of composite representation. The **exemplar view of concepts** makes just the opposite assumption: It asserts that concepts include representations of at least some actual individual instances. The exemplar approach assumes that people categorize new instances by comparing them to representations of previously stored instances, called *exemplars*. That is, people store representations of actual instances (Fido, the golden retriever with the long ears; Rover, the black and white sheltie who's missing a tail due to an unfortunate encounter

with a raccoon; Precious, the Yorkshire terrier who always has painted toenails and a bow in his hair).

Like the prototype view, it thus explains people's inability to state necessary and defining features: There are none to be stated. It also explains why people may have difficulty categorizing unclear, atypical instances: Such instances are similar to exemplars from different categories (for example, tomato is similar both to fruit exemplars, such as oranges or apples, and to vegetable exemplars, such as beets or squash) or are not similar enough to any known exemplars (Medin & Smith, 1984). Typical instances are thought to be more likely to be stored than less typical ones (Mervis, 1980) or to be more similar to stored exemplars, or both. This explains why people are faster to process information about typical instances. So, in trying to retrieve information about a typical instance, it is faster to find very similar stored exemplars. Atypical instances, in contrast, being rather dissimilar from stored exemplars, take longer to process.

The biggest problem with the exemplar view is that, like the prototype view, it is too unconstrained. It fails to specify, for example, which instances will eventually be stored as exemplars and which will not. It also does not explain how different exemplars are "called to mind" at the time of categorization. However, many psychologists believe people often store information about some specific category members in their conceptual representations, as we will see later.

The Schemata View

We have already touched on the concept of a schema, or organized framework for representing knowledge, in talking about Bartlett's (1932) work on people's memories for stories (see Chapters 6 and 7), so our coverage here will be brief. Be sure you've understood the material from those chapters on schemata as you read what is here.

The schemata view of concepts is that concepts are schemata—frameworks of knowledge that have roles, slots, variables, and so on. Schemata can embed themselves in one another hierarchically. Thus any schema can have subschemata and/or superschemata. The "meeting a college roommate for the first time" schema can be a subschema of an "orientation to college" schema, which may be embedded in an "attending college" schema, and so on. Similarly, the "meeting a college roommate for the first time" schema may contain subschemata of "dorm room," "meeting new person," and "roommate" embedded within it. The schema for "dog" may be a part of the schemata for "mammal," "pet," "animal," and "living thing"; it may contain subschemata such as "fur," "paws," and "wagging tail." The notion of schemata as underlying organizational units of memory has had significant impact on cognitive psychologists' thinking about how memory is organized and concepts represented. Some (Komatsu,

1992) have seen the **schemata/scripts view of concepts** as sharing features with both the prototype view (in that both schemata and prototypes store information that is abstracted across instances) and the exemplar view (in that both schemata and exemplars store information about actual instances).

The schemata view shares some of the problems facing the prototype and exemplar views. It does not specify clear enough boundaries among individual schemata. Moreover, some psychologists argue that in its current state the schema framework is not sufficiently delineated to be empirically testable (Horton & Mills, 1984). Answers to the following questions are still needed: What kinds of experiences lead to the formation of new schemata? How are schemata modified with experience? How do people know which schemata to call up in different situations—that is, what sorts of environmental cues are used?

The Knowledge-Based View

A number of cognitive psychologists (Keil, 1989; Lin & Murphy, 2001; Murphy & Medin, 1985) have argued that concepts have much more to do with people's knowledge and worldviews than previously recognized. Murphy and Medin (1985) suggested that the relationship between a concept and examples of the concept is analogous to the relationship between a theory and data supporting that theory. The idea of the **knowledge-based view of concepts** is that a person classifying objects and events doesn't just compare features or physical aspects of the objects and events to features or aspects of stored representations. Instead, the person uses his or her knowledge of how the concept is organized to justify the classification and to explain why certain instances happen to go together in the same category. The knowledge-based view helps explain how an apparently disparate collection of objects can form a coherent category in particular circumstances.

To take an example from Barsalou (1983), consider the category comprising children, pets, photo albums, family heirlooms, and cash. On the face of it, these things don't seem to go together very well, but in the context of a scenario in which a fire is about to engulf a house, these things fall neatly into the category "things to save." We know that each object mentioned is precious to its owner or parents and also irreplaceable. Notice, however, that the category becomes coherent only when we know its purpose.

Recall that the prototype, exemplar, and schemata/scripts approaches to concepts and categories fail to offer much of an answer to the question of how things in the same category go together. The knowledge-based view proposes that people's theories or mental explanations about the world are intertwined with their concepts and provide the basis for categorization (Heit, 1997). This view lets people explain to themselves and to others the instances that go

together and why, the features or aspects of instances that are important and why, and the features or aspects that are irrelevant and why.

The five approaches to conceptual structure just reviewed have been themselves categorized into two major types: similarity-based and explanation-based (Komatsu, 1992). The similarity-based category consists of the classical, prototype, and exemplar views (and some parts of the schemata/scripts view). It includes approaches in which categorization is assumed to be based on the similarity of an instance to some abstract specification of the category (such as a definition or a prototype) or to one or more stored exemplars.

However, to say that objects are categorized on the basis of similarity raises some problems, some of which Goodman (1972) pointed out. Consider two objects, a fork and a spoon. We say they are similar, probably because they share many properties: Both are made of metal, both are less than a foot long, and both are used as eating utensils. Now consider two other objects, a plum and a lawnmower. Are these similar? Well, they share several properties: Both weigh less than 100 kilos (and in fact, both weigh less than 101 kilos, 102 kilos, and so forth). In fact, these two apparently dissimilar items share an infinite number of properties (Hahn & Chater, 1997). But the property of weighing less than 100 kilos seems somehow beside the point when you are evaluating the similarity between a plum and a lawnmower. The key point is that similarity is meaningful only in certain respects. But Goodman concluded that the term *similarity* is pretty empty without some specification of what the relevant respects are.

Komatsu (1992) defined a different type of approach to concepts, which he called the *explanation-based category*, comprising some of the schemata/scripts view and some of the knowledge-based view. In this approach to the study of concepts, people are seen as basing classifications on meaningful relationships among instances and categories. The contrast between the similarity-based and the explanation-based approaches has to do with the degree to which people focus on superficial, perceptual information about a particular object versus the degree to which they focus on deeper, knowledge-derived information about an object's function or role.

The five approaches to concepts differ on several dimensions. The first dimension is the cognitive economy of the mental representation. Recall our discussion of cognitive economy from Chapter 7. The idea is to save on mental resources (such as storage space, processing time) by limiting the amount of information we must store. If we treated every single object or event as completely unique, thereby forming a unique mental representation for each, we would not be using our cognitive resources very economically.

In contrast, if we categorized all objects into one category (called "things"), the category wouldn't be very informative. So any theory of concepts and categorization must strike a balance between cognitive economy and informativeness (Komatsu, 1992). At the same time, any theory of concepts must explain

a concept or category's coherence—what holds the class of things together into a natural grouping. Some approaches, such as the classical approach, do this very directly; others have fuzzier boundaries around and between concepts.

Thus far, we have looked at different proposals for what concepts are—their nature and structure. In the next section, we will examine empirical studies looking at how concepts are actually formed and used. Presumably, understanding something about the ways in which people classify new instances can shed light on the nature of concepts. The studies we'll review next will help us think more carefully about the five approaches just discussed.

■ FORMING NEW CONCEPTS AND CLASSIFYING NEW INSTANCES

To have a concept of something is to group similar things together, to treat in more or less similar ways members of the category that the concept includes. To form a concept, people must have some basis for generalization, for grouping certain things but not others together. When you think about it, forming a concept is a remarkable cognitive achievement. It requires that we figure out which attributes or features of things are relevant and which should be ignored. Often we must carry out the task with very little feedback. In this section, we'll explore some investigations of how people manage this complex undertaking. Throughout, we'll see that psychologists' assumptions about how concepts are mentally represented have influenced their views on how people acquire or form new concepts.

Concept Attainment Strategies

Bruner, Goodnow, and Austin (1956) conducted some of the earliest work on how people form (or, in their terminology, “attain”) concepts. They saw several components in the process: acquiring the information necessary to isolate and learn a concept, retaining the information for later use, and transforming the information to make it usable when testing ideas about new possible instances.

Bruner et al. (1956) studied the ways people attained concepts, using cards depicting differing geometric figures, as shown in Figure 8-2. Note that each card has one of three shapes (circle, square, or cross), one of three colors (here, black, white, or striped), different numbers of shapes (one, two, or three), and different numbers of borders around the shapes (one, two, or three). The experimenter first placed before each participant all the cards appearing in Figure 8-2. Participants were told the experimenter had in mind a certain concept, such as “black circles” or “all cards containing two borders and striped figures.” Participants were then shown one card that illustrated the concept—in other